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THE MAIN QUESTION IS: CAN WE LEARN!

How can we optimally and efficiently estimate the function $f : X \rightarrow Y$ from a given set of data points $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, such that the trained model generalizes well to new, unseen data sampled from the same underlying distribution?

How do we efficiently "teach" or train a computer program (called a model) based on the **data** we provide? Just like different children might take varying amounts of time and examples to learn, different machine learning models might require different amounts of data and processing to train effectively.

THE MAIN QUESTION IN MACHINE LEARNING!

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K-NEAREST NEIGHBORS (K-NN) [REVISE THE TITLE

